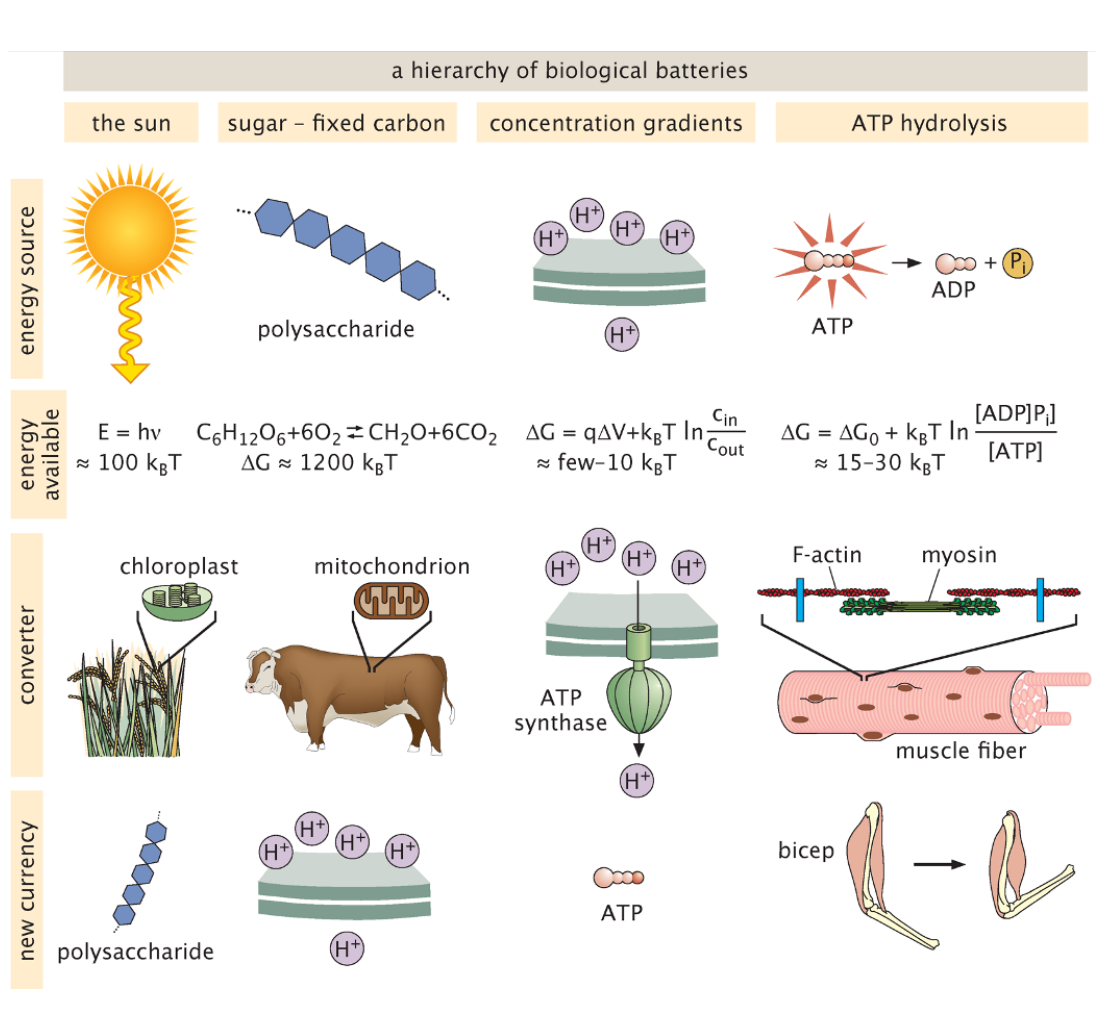
Lecture 17, Part II - ATP battery

Friday, March 11, 2022 6:11 AM



The ATP battery

$$ATP \rightleftarrows ADP + P$$

$$\Delta G = -\mu_{ATP} + \mu_{ADP} + \mu_P$$

$$= -\mu^0_{ATP} - k_BT\ln\left(\frac{c_{ATP}}{c_0}\right) + \mu^0_{ADP} + k_BT\ln\left(\frac{c_{ADP}}{c_0}\right) + \mu^0_P + k_BT\ln\left(\frac{c_P}{c_0}\right)$$

$$= \overbrace{\left(\mu^0_{ADP} + \mu^0_P - \mu^0_{ATP}\right)}^{\Delta G_0} + k_BT\ln\left(\frac{c_{ADP}}{c_0}\frac{c_P}{\cancel{c_0}}\frac{\cancel{c_0}}{c_{ATP}}\right)$$

$$= \Delta G_0 + k_BT\ln\left(\frac{c_{ADP}c_P}{c_{ATP}c_0}\right)$$

In equilibrium:

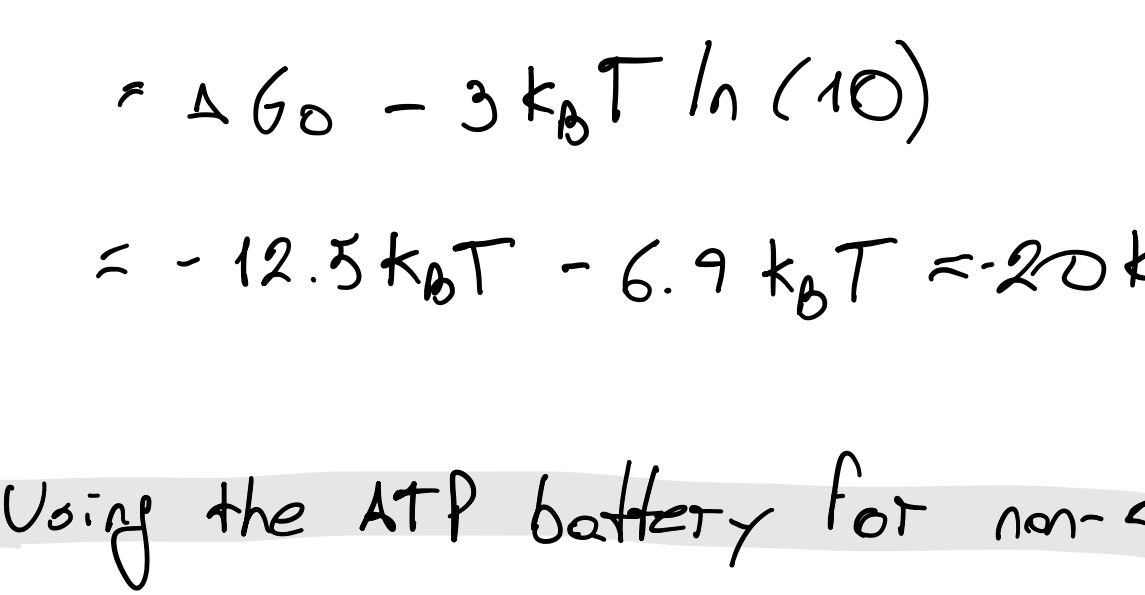
$$\Delta G = 0 = \Delta G_0 + k_BT\ln\left(\overbrace{\frac{c^{eq}_{ADP}c^{eq}_P}{c^{eq}_{ATP}}}^{K_D}\frac{1}{c_0}\right)$$

$$\Rightarrow \Delta G_0 = -k_BT\ln\left(K_D\frac{1}{c_0}\right)$$

$$= -12.5\ k_BT$$

↓ biochemical measurement

but in the cell



physiological condition of organism	ATP conc.	ADP conc.	P_i conc.	Inferred ΔG′ (kJ/mol)	Inferred ΔG′ (k_BT)	BNID
standard conditions	1 M	1 M	1 M	−36 to −38	−14 to −15	106580, ionic strength dependent
E. coli aerobic exponential growth on glucose	10 mM	0.6 mM	20 mM	−54	−22	104704
E. coli anaerobic exponential growth on glucose	3 mM	0.4 mM	10 mM	−54	−22	101964
E. coli aerobic exponential growth on glycerol	7 mM	0.7 mM	10 mM	−55	−22	101701
S. cerevisiae aerobic growth on glucose	2 mM	0.3 mM	22 mM	−52	−21	106017
spinach *Spinacia oleracea* chloroplast stroma in light	2 mM	0.8 mM	10 mM	−51	−21	108113
spinach *Spinacia oleracea* cytosol + mitochondria in light	3 mM	0.7 mM	10 mM	-52	−22	108113
spinach *Spinacia oleracea* cytosol + mitochondria in dark	1.5 mM	0.8 mM	10 mM	−50	−21	108113
Homo sapiens—resting muscle	8 mM	9 μM	4 mM	−68	−27	101943
Homo sapiens—muscle recovery from severe exercise	8 mM	7 μM	1 mM	−72	−29	101944

Table 3-2 Free energy for ATP hydrolysis in various organisms and under different physiological conditions. Inferred ΔG′ calculations based on a value of ΔG′° of −37.6 kJ/mol. This makes the values in the table consistent among themselves, but creates small deviations from the ΔG′ values reported in the primary sources. Such deviations can result from variations in ionic strength, pH, and biases in the measurement method. Values are rounded to one or two significant digits. In spinach, where P_i concentration was not reported, a characteristic value of 10 mM was used (BNID 103984, 103983, 111358, 105540).

$$\Delta G = \Delta G_0 + k_BT\ln\left(\frac{\cancel{20}\cdot\cancel{10^{-3}}\,\cancel{M}\cdot\cancel{0.6}\cdot10^{-3}\cancel{M}}{\cancel{10}\cdot\cancel{10^{-3}}\cancel{M}\cdot\cancel{1M}}\right)$$

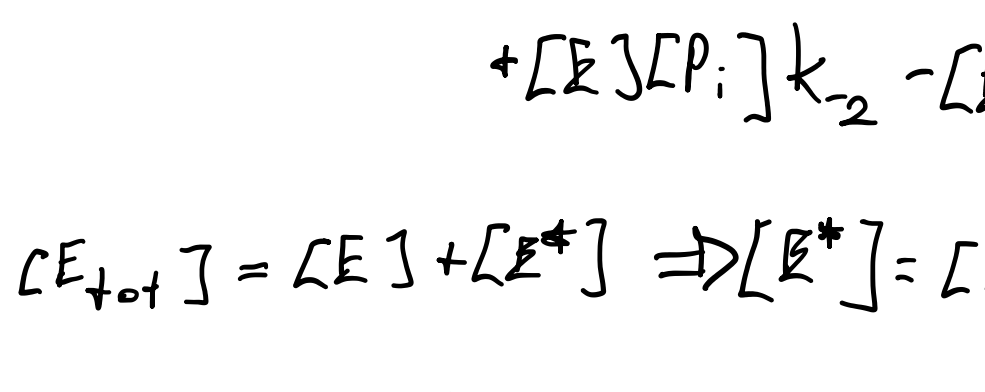
↓ E.coli

$$= \Delta G_0 + k_BT\ln\left(10^{-3}\right)$$

$$= \Delta G_0 - 3k_BT\ln(10)$$

$$= -12.5k_BT - 6.9\ k_BT \approx -20k_BT$$

Using the ATP battery for non-equil. steady states (and hypersensitivity)



$$E + ATP \underset{k_{-1}}{\overset{k_1}{\rightleftarrows}} E^* + ADP$$

$$E^* \underset{k_{-2}}{\overset{k_2}{\rightleftarrows}} E + P_i$$

The corresponding rate eqns.

$$\frac{d[E^*]}{dt} = -\frac{d[E]}{dt} = [E][ATP]k_1 - [E^*][ADP]k_{-1} + [E][P_i]k_{-2} - [E^*]k_2$$

$$[E_{tot}] = [E] + [E^*] \Rightarrow [E^*] = [E_{tot}] - [E]$$

In steady state

$$0 = [E][ATP]k_1 - [E^*][ADP]k_{-1} + [E][P_i]k_{-2} - [E^*]k_2$$

$$0 = [E]\cancel{[ATP]}k_1 - \left(\cancel{[E_{tot}]} - \cancel{[E]}\right)[ADP]k_{-1} + [E]\cancel{[P_i]}k_{-2} - \left(\cancel{[E_{tot}]} - \cancel{[E]}\right)k_2$$

$$0 = [E]\left(c_{ATP}k_1 + c_{ADP}k_{-1} + c_Pk_{-2} + k_2\right) - [E_{tot}]\left(c_{ADP}k_{-1} + k_2\right)$$

$$\Rightarrow [E] = [E_{tot}]\frac{c_{ADP}k_{-1} + k_2}{c_{ATP}k_1 + c_{ADP}k_{-1} + c_Pk_{-2} + k_2}$$

$$[E^*] = [E_{tot}]\frac{c_{ATP}k_1 + c_Pk_{-2}}{c_{ATP}k_1 + c_{ADP}k_{-1} + c_Pk_{-2} + k_2}$$

what are the fluxes through the system?

$$J_1^{net} = J_1^{E\to E^*} - J_1^{E^*\to E} = k_1[E]c_{ATP} - k_{-1}[E^*]c_{ADP}$$

$$J_2^{net} = J_2^{E^*\to E} - J_2^{E\to E^*} = k_2[E^*] - k_{-2}[E]c_P$$

In steady state, the net flux is zero:

$$J_1^{net} = J_2^{net}$$

In equilibrium, the condition is even stronger: each J_1/J_2 should be zero: detailed balance!

$$J_1 = 0 = k_1[E]c_{ATP} - k_{-1}[E^*]c_{ADP} =$$

$$= [E_{tot}]\left[k_1c_{ATP}\frac{c_{ADP}k_{-1} + k_2}{c_{ATP}k_1 + c_{ADP}k_{-1} + c_Pk_{-2} + k_2} - k_{-1}c_{ADP}\frac{c_{ATP}k_1 + c_Pk_{-2}}{c_{ATP}k_1 + c_{ADP}k_{-1} + c_Pk_{-2} + k_2}\right]$$

$$= [E_{tot}]\frac{\overset{!}{\underset{}{\overbrace{k_1c_{ATP}(c_{ADP}k_{-1} + k_2) - k_{-1}c_{ADP}(c_{ATP}k_1 + c_Pk_{-2})}^{0}}}}{c_{ATP}k_1 + c_{ADP}k_{-1} + c_Pk_{-2} + k_2}$$

$$\Downarrow$$

$$k_1c_{ATP}(c_{ADP}k_{-1} + k_2) = k_{-1}c_{ADP}(c_{ATP}k_1 + c_Pk_{-2})$$

$$\cancel{k_1c_{ATP}c_{ADP}k_{-1}} + k_1k_2c_{ATP} = \cancel{k_{-1}k_1c_{ADP}c_{ATP}} + k_{-1}k_{-2}c_{ADP}c_P$$

$$k_1k_2c_{ATP} = k_{-1}k_{-2}c_{ADP}c_P$$

$$\frac{k_1k_2}{k_{-1}k_{-2}} = \frac{c_{ADP}c_P}{c_{ATP}}$$

How much energy is being used per cycle

$$\Delta G_{\substack{one\\cycle}} = -\mu_{ATP} + \mu_{ADP} + \mu_P$$

$$= \overbrace{-\mu^0_{ATP} + \mu^0_{ADP} + \mu^0_P}^{\Delta G_0} - k_BT\ln\left(\frac{c_{ATP}}{c_0}\right) + k_BT\ln\left(\frac{c_{ADP}}{c_0}\right) + k_BT\ln\left(\frac{c_P}{c_0}\right)$$

$$= \underline{\Delta G_0} + k_BT\ln\left(\frac{c_{ADP}c_P}{c_{ATP}c_0}\right)$$

$$= -k_BT\ln\left(\frac{c^{eq}_{ADP}c^{eq}_P}{c^{eq}_{ATP}c_0}\right)$$

$$= -k_BT\ln\left(\frac{k_1k_2}{k_{-1}k_{-2}c_0}\right)$$

↓ from what we know about detailed balance

$$\Delta G_{\substack{one\\cycle}} = -k_BT\ln\left(\frac{k_1k_2}{k_{-1}k_{-2}c_0}\right) + k_BT\ln\left(\frac{c_{ADP}c_P}{c_{ATP}c_0}\right)$$

$$= k_BT\ln\left(\underbrace{\frac{c_{ADP}c_P}{c_{ATP}}\cdot\frac{k_{-1}k_{-2}}{k_1k_2}}_{=1\text{ in equil}}\right)$$

$$\Rightarrow \Delta G_{\substack{one\\cycle}} = 0$$

Fraction of phosphorylated enzyme in steady state: a toy ATP sensor

$$f = \frac{[E^*]^{ss}}{[E^*]^{ss} + [E]^{ss}} = \frac{k_1c_{ATP} + k_{-2}c_P}{k_1c_{ATP} + k_{-2}c_P + k_{-1}c_{ADP} + k_2}$$

Detailed balance condition

$$\frac{c_{ADP}c_P}{c_{ATP}} = \frac{k_1k_2}{k_{-1}k_{-2}} \Rightarrow c_{ADP} = \frac{c_{ATP}}{c_P}\frac{k_1k_2}{k_{-1}k_{-2}}$$

$$f = [E_{tot}]\cdot\frac{k_1c_{ATP} + k_{-2}c_P}{k_1c_{ATP} + k_{-2}c_P + \cancel{k_{-1}}\frac{c_{ATP}}{c_P}\frac{k_1k_2}{\cancel{k_{-1}}k_{-2}} + k_2}$$

$$= [E_{tot}]\cdot\frac{k_1c_{ATP}c_P + k_{-2}c_P^2}{k_1c_{ATP}c_P + k_{-2}c_P^2 + c_{ATP}\frac{k_1k_2}{k_{-2}} + k_2c_P}$$

$$= [E_{tot}]\frac{k_1k_{-2}c_{ATP}c_P + k_{-2}^2c_P^2}{k_1k_{-2}c_{ATP}c_P + k_{-2}^2c_P^2 + c_{ATP}k_1k_2 + k_2k_{-2}c_P}$$

$$= [E_{tot}] = \frac{(k_1c_{ATP} + k_{-2}c_P)k_{-2}c_P}{(k_1c_{ATP} + k_{-2}c_P)k_{-2}c_P + k_1k_2c_{ATP} + k_2k_{-2}c_P}$$

$$= \frac{k_{-2}c_P}{k_{-2}c_P + \frac{k_1k_2c_{ATP} + k_2k_{-2}c_P}{k_1c_{ATP} + k_{-2}c_P}}$$

$$= \frac{k_{-2}c_P}{k_{-2}c_P + k_2\cancel{\frac{k_1c_{ATP} + k_{-2}c_P}{k_1c_{ATP} + k_{-2}c_P}}}$$

$$= \frac{k_{-2}c_P}{k_{-2}c_P + k_2}$$

without imposing detailed balance

$$f = \frac{[E^*]^{ss}}{[E^*]^{ss} + [E]^{ss}} = \frac{k_1c_{ATP} + k_{-2}c_P}{k_1c_{ATP} + k_{-2}c_P + k_{-1}c_{ADP} + k_2}$$

$$\frac{\overset{\theta}{\frac{k_1c_{ATP}}{k_2}} + \overset{\mu}{\frac{k_{-2}c_P}{k_2}}}{\overset{\theta}{\frac{k_1c_{ATP}}{k_2}} + \underset{\mu}{\frac{k_{-2}c_P}{k_2}} + \underbrace{\frac{k_{-1}c_{ADP}}{k_2}} + 1}$$

$$\Downarrow$$

$$\frac{k_{-1}c_{ADP}}{k_2}\cdot\frac{\overset{\theta}{k_1c_{ATP}}}{k_1c_{ATP}} =$$

$$= \theta\frac{k_{-1}c_{ADP}}{k_1c_{ATP}}\frac{k_{-2}c_P}{k_2}\overbrace{\frac{k_2}{k_{-2}c_P}}^{1/\mu}$$

$$= \frac{\theta}{\mu}\frac{k_{-1}k_{-2}}{k_1k_2}\cdot\frac{c_{ADP}c_P}{c_{ATP}}$$

Remember that

$$\Delta G_{\substack{one\\cycle}} = k_BT\ln\left(\frac{k_1k_2c_{ATP}}{k_{-1}k_{-2}c_{ADP}c_P}\right)$$

$$\frac{k_1k_2}{k_{-1}k_{-2}}\cdot\frac{c_{ATP}}{c_{ADP}c_P} = e^{\Delta G/k_BT} = \gamma$$

$$\Rightarrow f = \frac{\theta + \mu}{1 + \theta + \mu + \frac{\theta}{\mu\gamma}}$$

$$f = \frac{[E^*]}{[E] + [E^*]} = \frac{\theta + \mu}{1 + \theta + \mu + \theta/(\gamma\mu)},$$

e $\theta = k_1/k_2$, $\mu = k_{-2}/k_2$, and $\gamma = k_1k_2/(k_{-1}k_{-2}) = \exp(\Delta\mu_{DT}/k_BT)$.

$\gamma = 1$ (no energy)

$$= \frac{\theta + \mu}{1 + \theta + \mu + \theta/\mu}\cdot\frac{\mu}{\mu} = \frac{\mu\theta + \mu^2}{\mu + \theta\mu + \mu^2 + \theta}\cdot\frac{\frac{1}{\theta+\mu}}{\frac{1}{\theta+\mu}}$$

$$= \frac{\mu}{\overset{1}{\cancel{\mu + \theta}} + \mu(\cancel{\mu + \theta})}\cdot\frac{1}{\cancel{\frac{1}{\theta + \mu}}} = \frac{\mu}{1 + \mu}$$